

Lecture 11-18

Der Mann Ohne Eigenschaft

Linear Operator : Evolution
(Perron-Frobenius)

$$P(x, t) = (\mathcal{L}^t \circ \mathcal{P})(x) = \sum_{x_0 \in \mathcal{M}} \frac{P(x_0)}{|\det \mathcal{J}^t(x_0)|}$$

$x_0 \in \mathcal{M}$

$$\mathcal{L}^t(y, x) = \delta(y - F^t(x))$$

Matrix

$$P(x, t) = \int_{\mathcal{M}} dx \mathcal{L}^t(y, x) P(x)$$



- We have given up on taking a 'well defined' IC for a deterministic trajectory
- However, for non-integrable, chaotic systems, you can wind up anywhere
- after one Thurston (hyperbolic time), system can be anywhere!
- start from ground up: distributions of IC's!

→ Perron Frobenius Operator admits (semi-group) infinite group properties:

$$(a) \mathcal{L}^{t=0} = \mathbb{1}$$

$$(b) \mathcal{L}^t \circ \mathcal{L}^{t'} = \mathcal{L}^{t+t'}$$

$t, t' \geq 0$ (semi-group only lets $t \geq 0$)

Chaos Book Sec. 19.4: Invariant Measures

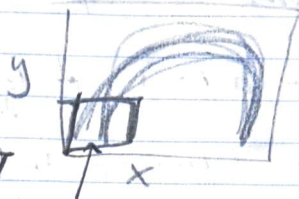
- Examples are fixed points, limit cycles, and strange attractors of higher dimension
- However, the utility of invariant measures varies, since those that are point-line or unstable are not robust to being perturbed
- We then introduce Natural Invariant Measures

Natural Invariant Measures (Stationary)

$$(\mathcal{L}^t \circ \rho_0)(x) = \mathbb{1} \cdot \rho_0(x)$$

- The eigenvalue problem appears when we consider the fixed points (invariant measures) of the PF operator
- Remember back to undergrad: The Lyapunov spectrum (calculated from this eigenvalue problem using the PF operator) dictates the behavior of n -dimensional chaotic systems

→ We can also compute this numerically
invariant measure



Strange Manifold of Lorenz
projected onto $z=0$

→ can compute an average # of visits per orbit to
each part of this region

→ but this isn't the most effective method

→ consider averaging over time

$$\overline{p(y)} = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t \delta(y - f^z(x)) dz$$

→ But, there exist points/orbits, that if we
started on them, they wouldn't describe the
'generic' behavior of the system, because they
are fixed points or limit cycles. Then, the
value is parameterized by / a function of
the starting point:

$$\overline{p}_x(y) \quad \text{generic starting point}$$

$\Delta t_i / \overline{t} = \frac{t_i}{t}$ → but if you try this in higher
dimensions, it breaks down b/c
Most points are never reached
dwell/visit time

Expectation Values

(similar to QM)

→ Consider a topographic map, i.e., an observable

$$a = a(x) \leftarrow \text{local elevation in Atlanta, e.g.}$$

→ Say we want to determine the mean height
→ but we can't survey the entire city of ATL.
So we will spend lots of times in some areas
and little time in others

↳ We need measure theory machinery to account for this

$$\text{Space avg: } \langle a \rangle = \int_M dx \, p(x) a(x)$$

The $p(x)$ represents the density function, accounting for where we do survey and the time density of it.

→ Then, $d\mu \equiv dx \, p(x)$ is the measure

$$\Rightarrow \langle a \rangle_p = \int_M d\mu \, a(x)$$

($d\mu$ is natural measure)

$$\text{time avg: } \bar{a}_{x_0} = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t ds \, a(f^s(x_0))$$

→ these averages are the same in most cases! → Birkhoff Theorem
(Ergodic Theory)

space $\langle a \rangle_p = \int_M dx p(x) a(x) = \int_M d\mu a(x)$

time $\bar{a}_x = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t dz a(f^z(x_0))$

space-time $\langle a \rangle = \frac{1}{|M|} \int_M \bar{a}_x dx$

→ nonlinearity is gone!

→ everything is the same! QM, chaos, stock markets, they all just have very different kernels

field	QM	classical	Stochastic
kernel	$e^{i\hbar}$	dirac δ	Gaussians
behavior	spreads out/ distributions	very hard/ rigid, symplectic space	erratic, predictable